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RESEARCH-ARTICLE

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Abstract

NodeGrade tries to provide a suitable solution for the problem of time-intensive short answer grading. This research focuses simultaneously on performance, functionality and user experience, which is underlined by a triangulated approach. The evaluation results show comparable performance of NodeGrade on public datasets, even outperforming GPT-4 on the SemEval 2013 Task 7. Matching of NodeGrade's output with multiple human expert raters reveals some weaknesses regarding cases at the lower and upper boundary. In terms of user experience, the interviewed and observed students recognized both positive facets, like better learning support and helpful feedback, and negative sides, including technical limitations and lack of transparency. Overall, NodeGrade promises high potential for further practical use and testing in the field of software engineering education and automatic short answer grading.

CCS Concepts

• **Computing methodologies** → **Natural language processing**;
• **Applied computing** → *Interactive learning environments.*

Keywords

ASAG, Automatic Short Answer Grading, Short Answer Scoring, AI in Education, Software Engineering Education, Natural Language Processing, Large Language Models

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1 Introduction

Although the full potential of artificial intelligence as an emerging field remains to be explored, it has already become a significant aspect of modern education with a wide range of applications. One such application is adaptive learning systems, which aim to shape the individual learning process to students' specific needs. A key element of these systems is the provision of tailored support and assessments regarding student tasks [22]. In particular, free-text and essay-style questions yield students responses that can be individually assessed, thereby offering opportunities for personalized feedback and enhanced outcomes [14] in software engineering education. However, these benefits are counterbalanced by the considerably increased time required for grading.

Automatic short answer grading (ASAG) has been the focus of extensive research over many years [12] as a potential solution to the challenges associated with grading free-text responses. It was identified by [10] that recent studies have introduced novel approaches and applied the emerging trend of large language models (LLMs) to the domain of ASAG. Fischer et al. [11] proposed a flexible approach that employs a node graph, which is fully customizable and adaptable to specific short answers. The goal was to provide an alternative to the prevalent one-size-fits-all solutions that offer a single toolset for various domains and applications in grading and scoring short answers. Moreover, the work offers a means for the toolset to deliver customized textual feedback to learners. The node graph permits the integration of various natural language processing (NLP) techniques, depending on the specific use case and user, and seeks to circumvent several issues associated with manual grading, such as scalability, consistency, and the absence of personalized feedback [10]. To evaluate the effectiveness of NodeGrade (NG) as a node-based ASAG tool, this study addresses the following research questions:

- RQ1: How does NG perform on public datasets when evaluated using the QWK metric?
- RQ1.1: How does NG compare to existing ASAG systems under common metrics?
- RQ2: How does NG's scoring align with human grading of multiple experts on selected tasks in software engineering education?
- RQ3: How do software engineering students interact with NG, what are their perceptions of its use, and what improvements do they suggest?

RQ1 evaluates NG's performance on standardized datasets, assessing its ability to generalize across different domains and question types. RQ1.1 compares NG's performance to state-of-the-art (SOTA) ASAG systems, highlighting its strengths and limitations relative to current approaches. RQ2 examines the degree to which NG's automated scoring matches the judgments of multiple human experts, providing insights into its reliability and potential as a grading tool. RQ3 focuses on the opinions and interactions of students regarding NG to gain feedback and potentially optimize its use for software engineering education.

The structure of the paper is as follows: Chapter 2 presents related work on ASAG, assessment approaches, LLM integration, fitting public data and the evaluation of educational technologies. Chapter 3 introduces the research methods applied in the evaluation of NG including the experimental settings, quantitative analysis and qualitative research procedures. Chapter 4 gives an overview of the evaluation results that get discussed, interpreted and put into the context of the collected dataset in Chapter 5. Limitations of these findings are also outlined. Finally, the conclusions along with prospects for future work are presented in Chapter 6.

2 Related work

2.1 Field Overview & Evolution

The field of ASAG has demonstrated variations in disciplinary focus since its inception. Zawacki-Richter et al. [29] observed that a large proportion of ASAG research emphasizes STEM areas, which was quantified by Gao et al. [12] as 55% of analyzed studies, with Computer Science alone accounting for 45%. These tendencies are partly influenced by the structured nature of STEM responses and the historical availability of specialized datasets.

Gao et al. [12] provide a taxonomy for ASAG systems based on two major dimensions: system output types (classification, grading, and feedback generation) and input modalities (textual, multimodal). Their findings outline an evolution from classical NLP methods that use deterministic algorithms to modern techniques that incorporate transformer architectures and semantic similarity metrics. Fagbohun et al. [10] highlight this progression, indicating a transition from traditional feature engineering toward approaches based on LLMs.

2.2 Assessment Approaches

Early ASAG systems relied primarily on classical NLP methods, including feature extraction based on syntactic patterns, lexical overlap, and rule-based techniques. These methods often integrated human intervention, where automated scoring was supplemented by educator review. Common practices involved bag-of-words representations, TF-IDF weighting, and semantic similarity measures using WordNet or latent semantic analysis.

Subsequent work introduced more advanced approaches, with feature engineering shifting from manual design to automated representation learning. Transformer-based models such as BERT and RoBERTa offered better contextual understanding, while hybrid systems combined statistical NLP with knowledge graphs for domain-specific evaluations. However, as Gao et al. [12] indicate, 51% of the studies still focus on optimizing performance metrics rather than integrating pedagogical goals.

The adoption of LLMs marked an important stage in this field, especially for open-ended responses and more complex reasoning. Early systems typically reported 70–80% accuracy on limited datasets, whereas modern approaches benefit from pre-trained models refined with educational content. This direction aligns with the focus identified by Gao et al. [12], where improving model performance (51%) outpaces objectives like teacher support (13%) or student feedback (12%). Although the overall direction parallels broader NLP developments, it raises concerns about educational validity, which will be addressed in subsequent sections.

2.3 LLM Integration

LLMs have broadened the scope of ASAG systems, offering expanded capabilities with associated trade-offs. Kortemeyer et al. [18] shows that general-purpose LLMs such as GPT-4 achieve performance on SciEntsBank (F1 = 0.731) comparable to specialized systems (e.g., RoBERTa) fine-tuned on educational data, though a 12–15% performance gap remains for reasoning tasks. This gap narrows when using reference-free assessment methods, where GPT-4 demonstrates effective grading (F1 = 0.651 on Beetle) without solution exemplars, which supports formative feedback use cases.

Recent work emphasizes the role of prompt engineering in LLM-based ASAG. Chu et al. [5] report that automated prompt optimization and chain-of-thought techniques increase grading accuracy by 18% compared to zero-shot methods, though questions about reproducibility arise due to their unpublished dataset. Practical implementations must also consider changing API costs, data privacy, and computational demands highlighted by Fagbohun et al. [10]. Local alternatives (e.g., Llama 2) attain 68% of GPT-4's performance on reference-free tasks [18], while many cost-limited applications rely on GPT-3.5, which has shown 22% lower consistency scores [3]. The choice between embedding-based and completion-based models reflects different strengths. Ta Duong et al. [7] identified that an embedding-based approach (text-embedding-ada-002) with cosine similarity measurement achieved higher Pearson's r of 0.694 for software engineering assessments, while completion-based systems with GPT-3.5 enabled more nuanced feedback (Pearson's r = 0.525). Retrieval augmented generation methods further enhance context-aware grading by 31% when supplemented with course materials. However, these approaches are less feasible for unknown domains or in benchmarks like Beetle, where supplementary materials or imagery supporting the questions are unavailable.

The authors in [27] use LLMs to iterate on predefined rubrics for students answers. Rubrics are structured grading criteria including but not limited to expectations and point distribution. They contributed a systematic approach for grading using LLMs. Their results highlight the importance of prompting in the context of grading using LLMs.

2.4 Datasets & Evaluation

Benchmarking in ASAG frequently uses datasets like Mohler et al. [24], which contains 2,273 computer science answers (scored 0–5), and domain-general resources such as SciEntsBank and Beetle, referenced in 68% of comparative studies [18]. Although these datasets allow direct performance comparisons, Mohler relies on graduate-student raters with potential expertise bias, and Beetle focuses on fundamental science topics with limited difficulty.

More recent datasets address broader coverage and detailed annotation. Aggarwal's EngSAF [1] covers 5,800 responses from 25 engineering courses, includes three-tier human labels, and provides LLM-generated feedback. This synthetic feedback process may introduce biases, but the dataset's label distribution (12%–45%–43% for scores 0–1–2) and emotional impact analysis expand evaluation criteria. Human annotators evaluate this feedback with an average of 4.55 out of 5, indicating a reliability and quality.

Additional domain-specific corpora are emerging to support specialized tasks [7]. Ta Duong's software engineering dataset permits systematic comparisons of embedding-based and completion-based methods, while Chang Li-Hsin's [3] Finnish corpus highlights cross-linguistic considerations. GPT-4 scores 0.6 Relative Merit Consensus on 59% of questions in this corpus despite training on 24k exam responses, revealing areas for further investigation. Metrics in ASAG remain predominantly quantitative (52% benchmark-focused) [12].

2.5 Evaluation of educational technologies

When thinking about integrating a software into the context of higher education there is a necessity of evaluating the technology from the perspective of the most relevant stakeholders. The process is vital for determining whether the educational software improves learning and its reasons for success or failure. There are a variety of approaches to this topic and the literature focuses on diverse factors. For example Escudeiro et al. [9] built the Quantitative Evaluation Framework that included three requirement dimensions. Functionality being the first dimension is focusing on ease of use and the content's quality. The second dimension, adaptability, is among other things based on versatility, stimulation of initiative and self learning as well as the available didactic resources, meaning different activity types and tutoring options. The audiovisual quality, navigation and interaction are some of the representing factors for the third dimension of efficiency [9]. Another example would be Karolcik et al. [30] who have designed the Comprehensive Evaluation of Electronic Learning Tools and Educational Software (CEELTES) concept to evaluate the quality of educational software. It is based on five key requirements: stable and high-comfort technology, thorough and high-quality digital content considering pedagogical and psychological aspects of learning, simple and intuitive use, interactive digital content providing feedback and supplement material as well as reasonable use [30]. Our evaluation study is drawing inspiration from both of these approaches and focusing on the usability and functionality and efficiency of NG.

3 Research methods

We decided on a mixed methods approach, specifically data triangulation [25]. Therefore the methodology includes different quantitative and qualitative research procedures, which will be described in the following. We utilize public datasets to measure NG's performance quantitatively. After that, we compare it with other works in the field both addressing RQ1 and RQ1.1. Furthermore, the research team enacted multiple expert ratings of the students' solution that provided two of three feedback mechanisms of NG, namely an estimate of the score and whether or not the solution can be judged as adequate, for comparative measures regarding RQ2. Comparing the verbal feedback is beyond the scope of the current paper. Lastly, for the purpose of gathering student feedback and dealing with RQ3, the research team conducted observations and guided interviews with six students about NG.

3.1 Experimental Settings

Our experimental design focuses on evaluating a minimal yet effective node graph configuration for ASAG tasks[1], combining semantic similarity analysis with LLM-based assessment. The architecture and workflow are illustrated in [11].

3.1.1 Node Graph Setup. The graph employs a dual-path evaluation system similar to Ta Duong et al.'s [7] hybrid approaches, combining:

(1) Semantic similarity analysis using sentence transformers
(2) LLM-based scoring through carefully constrained prompts

This minimal configuration intentionally excludes additional preprocessing or feature engineering components to isolate the core assessment mechanisms. The three primary inputs (question, reference answer, and student answer) follow identical preprocessing:

- Trimming leading/trailing whitespace
- Removing consecutive spaces

We consciously avoid stemming or lexical normalization as it can lead to over-consolidation which alters semantic meaning [2, 28].

3.1.2 Prompt Design & Localization. The prompt structure was optimized through iterative testing to ensure consistent output formatting while minimizing engineering bias:

- System message:
```
You are an e-rater assessment system.
You reply truthfully and correctly.
```
- Task prompt:
```
 Compare the student answer with the sample
solution and output a value between 0 and 100.
Answer exactly in this format: "score: <number>".
```

For German-language evaluations, the prompts were translated using DeepL (version Jan. 2025) and manually validated by the first author. Benchmark comparisons used direct English translations

[1]Graph configuration used in experiments: https://github.com/HASKI-RAK/paper/tree/main/NodeGradeEvaluation/benchmark/graphs

to control for language performance disparities in CommonCrawl-trained models [20].

Table 1: Model and Parameters

Component	Details
LLM	CalmeRys-78B-Orpo-v0.1[2]
Parameters	
Temperature	0
Max tokens	200
Top_p	0.85
Top_k	50
Repetition penalty	0
Presence penalty	0
Repetition penalty range	512
Guidance scale	1.1
Sentence Transformer	all-mpnet-base-v2

3.1.3 Model Configuration. The parameter choices listed in Table 1 follow OpenAI API defaults unless otherwise specified, as comprehensive studies on ASAG-specific parameter tuning remain lacking (see Chapter 6). We increased the guidance scale as we noticed the model more often follows the given output format. We noticed that neither tweaking top_p nor top_k had a huge impact on the output since the model was prompted to output a single word, colon, whitespace and number.

3.1.4 Score Calculation. The final score ($score_{final}$) combines LLM assessment ($score_{LLM}$) and semantic similarity ($cosineSimilarity$) through simple averaging:

$$score_{final} = \frac{score_{LLM} + 100 \cdot cosineSimilarity(a, b)}{2}$$

where a and b are sentence transformer embeddings of reference answer and student answer. The 0-100 scaling aligns both components' ranges while preserving interpretability against ordinal benchmarks. This equal weighting serves as a pragmatic baseline pending future optimization studies mentioned in Chapter 6.

3.1.5 Implementation Details. Experiments ran on the provided hardware of 4×NVIDIA RTX A6000 GPUs (48GB VRAM each). This is sufficient to run a 78B parameter SOTA transformer model.

3.2 Datasets and quantitative analyses

To evaluate NG in relation to RQ1, we utilize publicly available datasets commonly referenced in the literature.

The EngSAF dataset [1] was obtained from the authors' public GitHub repository[3]. EngSAF comprises 5,814 student responses to 119 questions across 25 engineering courses. The grading labels follow an ordinal scale (0, 1, 2). The dataset consists of six columns: question ID, question text, reference answer, student response, grading label, and textual feedback. It is divided into train and test splits, with the test split further categorized into unseen answers (UA) and unseen questions (UQ). As the test split is not publicly available, our analysis is restricted to the train split.

Additionally, we use the SciEntsBank and Beetle datasets, which were featured in SemEval 2013 Task 7 [8], sometimes also referred to as Student Response Analysis (SRA) dataset. SciEntsBank contains 197 questions from 15 scientific domains. Its test split has approximately 11,678 student responses, each accompanied by a single reference answer. Beetle's test split comprises 11,978 responses to 56 questions, with six different reference answers grouped by response quality: best, good, minimal, or no classification. These datasets were acquired from Hugging Face, as the original distribution is no longer available. The Hugging Face version was compiled from various sources, including Kaggle and GitHub, by Atomi.[4].

SciEntsBank provides 3-way labels (correct, incorrect, contradicting), while Beetle uses a 5-way classification (correct, contradicting, partially correct, non-domain, irrelevant). Both datasets are split into train and test sets, but only the test split is used in our study.

Finally, we employ the OS dataset [27], which was collected from an undergraduate operating systems course at the City University of Hong Kong. This dataset consists of six questions supplemented with contextual information related to programming tasks. The grading scale varies by question, with scores ranging from 0 to 40. The final score is determined by the average of three independent human ratings. The dataset was acquired from the authors' GitHub repository[5].

To address RQ1we instructed NG to assign a correctness score to all student responses in the aforementioned datasets. This score, as described in chapter 3.1, is an interval-scaled decimal value ranging from 0 to 100. However, none of the reference datasets provide purely interval-scaled labels. While the Mohler and OS dataset features a quasi-interval-scaled evaluation score, the EngSAF and SemEval datasets only provide ordinal-scaled target variables. We addressed this issue by linearly transforming the model score to match the range of the respective dataset. For example, we mapped the score to the 0–5 range for Mohler and the 0–2 range for EngSAF. Subsequently, we rounded the transformed score to the nearest value present in the original scale. Through this process, the interval-scaled NG output was discretized into a categorical format, ensuring alignment with the predefined classification scheme of the respective dataset. This transformation enables the use of Quadratic Weighted Cohen's Kappa (QWK) [6], a metric suited for ordinal classification. However, since discretization inherently leads to information loss and may negatively impact prediction performance, we retained the transformed but non-discretized model scores for supplementary analyses. Hence, we applied additional measures tailored to the nature of each dataset. For datasets with interval-scaled reference scores (Mohler and OS), we computed Pearson's correlation coefficient to quantify the linear relationship between NG's raw outputs and ground-truth labels. Additionally, we visualized the relationship between continuous model scores and ordinal reference values using actual vs. fitted visualizations, including scatterplots and heatmaps. For datasets with strictly categorical annotations (EngSAF: three categories; SemEval: two categories),

[2]Top-ranked on HuggingFace Open LLM Leaderboard on evaluation metrics: IFEval, BBH, MATH, GPQA, MUSR, MMLU-PRO at the time of writing this work: https://huggingface.co/spaces/open-llm-leaderboard/open_llm_leaderboard
[3]https://github.com/dishankaggarwal/EngSAF
[4]https://www.getatomi.com/about
[5]https://github.com/wenjing1170/llm_grader

we measured NG's discriminative power using ROC-AUC. This metric assesses the model's ability to differentiate between response categories without being affected by discretization or thresholding constraints. By combining QWK with Pearson's r, ROC-AUC and actual vs. fitted visualization techniques, we provide a comprehensive view on NG's performance across different datasets.

3.3 Participants and qualitative research procedures

Expert ratings were given by three professors and a research assistant, all experienced both in the field of software engineering and in giving students feedback on their software engineering exercises. The raters were provided with the task, the student answers and the example solution that represented a perfect score. Based on this information they were asked to score the student answer out of 100 points and judge whether the answer passed or failed the task. They had no information about the score given by NG as to avoid a bias. All of the six students were enrolled in the software engineering course while being in their third semester of the game engineering study program of the Kempten University of Applied Sciences. They were recruited in pairs from the designated lecture and participated voluntarily in exchange for a small financial incentive. The research was integrated in a broader evaluation of an underlying adaptive learning system with a duration of about two and a half hours of observations, during which two students each worked in tandem on the topic of design patterns, and 20 minutes of retrospective interviews. An office space was set up for the selected student pairs and their designated timeslots, providing a quiet environment for undisturbed work. Participants received detailed information about the observation process and the related data protection measures. The research team streamed both screen recordings and camera footage to an adjacent office, intervening only in the event of technical issues. Throughout the observation, the team recorded any notable occurrences or challenges faced by the students. These observations were later discussed in a retrospective interview, that included the students in pairs as well as two researches and was guided by a pre-established framework. The guide was constructed on the basis of deductive categories that were expected to be relevant for the student users. It included interview questions regarding NG's usability, feedback mechanisms, design and perceived influence on the learning process. To analyze the students' statements about NG, a qualitative content analysis (QCA) was conducted to develop a category system and assign individual codes to main and subcategories. This QCA followed Mayring's model [23], employing a deductive-inductive approach to form the categories. This method involved creating categories both prior to and during the material review.

4 Results

4.1 Performance on public datasets

Table 2 displays NG's performance on the four publicly available datasets. For Mohler, where interval-scaled ground-truth scores are available, Pearson's correlation coefficient and QWK are reported. Both the English and German prompts yield identical correlation values (0.73) on Mohler dataset with only marginal differences in QWK (0.55 vs. 0.54). This indicates consistent performance across the two language conditions. For SemEval and EngSAF, the target variables are ordinal and strictly categorical, respectively. Hence, QWK and ROC-AUC are the primary performance indicators. On the SemEval dataset (considered as a whole), the QWK of 0.44 and ROC-AUC of 0.81 suggest moderate agreement with the ground-truth labels and good ability to distinguish between answer categories. When splitting SemEval into its "Beetle" (QWK: 0.40, ROC-AUC: 0.77) and "SciEntsBank" (QWK: 0.49, ROC-AUC: 0.85) subsets, performance varies, with the SciEntsBank subset showing stronger discrimination. EngSAF exhibits a QWK of 0.59 alongside a weighted one-versus-all ROC-AUC of 0.81 (calculated as the class-frequency-weighted sum of per-class ROC-AUC values). This indicates that NodeGrade achieves a balanced trade-off between agreement with the ordinal scale and the theoretical ability to identify different correctness levels. Table 3 presents an item-wise breakdown of performance on the OS dataset, which provides interval-scaled reference scores for six different questions. Overall, Pearson's correlation coefficient and QWK vary substantially across items. For both english and german prompts, Question 3 achieves the highest correlation (0.70), accompanied by the largest QWK values (0.68+). By contrast, while Question 1 performs poorly under the English condition, the German prompt results indicate that Question 6 is the most challenging item in that setting.

Table 2: Performance Metrics

Dataset	r	QWK	ROC-AUC
Mohler (English prompts)	0.73	0.55	n.a.
Mohler (German prompts)	0.73	0.54	n.a.
SemEval (whole dataset)	n.a.	0.44	0.81
SemEval (Beetle)	n.a.	0.40	0.77
SemEval (SciEntsBank)	n.a.	0.49	0.85
EngSAF	n.a.	0.59	0.81*

*r = Pearson's correlation coefficient, QWK = Quadratic Weighted Kappa, ROC-AUC = Receiver Operating Characteristic - Area under the Curve, n.a. = not applicable, * Weighted One-vs-all ROC-AUC*

Table 3: Performance Metrics on OS Dataset for English and German Prompts

Question-ID	English Prompts		German Prompts	
	r	QWK	r	QWK
3	0.70	0.69	0.70	0.68
2	0.46	0.44	0.46	0.44
4	0.42	0.37	0.55	0.49
5	0.60	0.35	0.62	0.39
6	0.23	0.20	0.22	0.21
1	0.23	0.20	0.51	0.35

r = Pearson's correlation coefficient, QWK = Quadratic Weighted Kappa.

Figure 1 presents the actual-vs-fitted heatmap for Mohler (utilizing English prompts)[6]. Each cell represents the percentage of

[6]The same visualization utilizing German prompts can be found on https://github.com/HASKI-RAK/paper/blob/main/NodeGradeEvaluation/supplementary_graphs.pdf but is omitted here due to negligible differences.

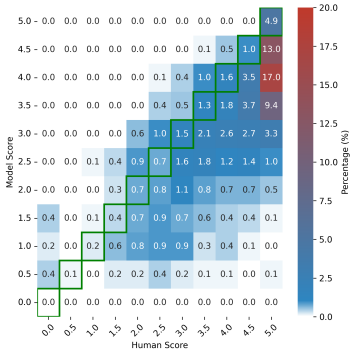

Figure 1: Actual-vs-Fitted Heatmap (Mohler Datset, English prompts)

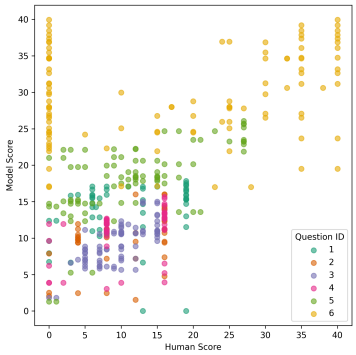

Figure 2: Actual-vs-Fitted Scatterplot (OS Dataset, English prompts, grouped by Question-ID)

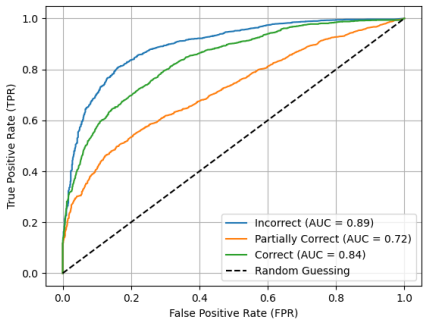

Figure 3: One-vs-all ROC Curves (EngSAF Dataset)

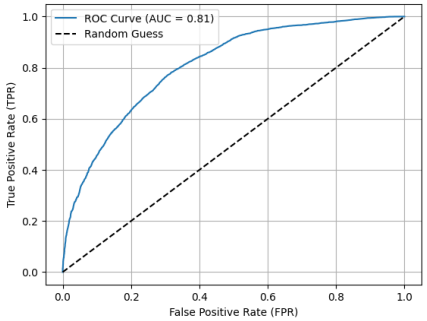

Figure 4: ROC Curve (SemEval Dataset)

responses that fall into a specific combination of human-assigned (horizontal axis) and model-predicted (vertical axis) scores on the dataset-specific 0-5 scale. While a concentration of cells around the main diagonal suggests some alignment between NG's predictions and ground-truth scores, notable deviations can be observed. In particular, for human scores above approximately 1.5, NG tends to assign lower scores compared to human raters in the reference dataset. Figure 2 presents the performance of NG on the OS dataset utilizing English Prompts[6]. Each dot represents a student response, with the horizontal axis showing the human-assigned score (ranging up to 40) and the vertical axis showing the score assigned by NG (linearly scaled to the same range). Points are color-coded by question ID. A general upward trend is evident, reflecting the positive correlation values reported in Table 3. However, the dispersion around the diagonal varies considerably by item. For instance, responses from question 3 cluster more tightly around the diagonal, consistent with its high correlation (r = 0.70). By contrast, question 6 exhibits a broader spread, aligning with its lower correlation and indicating that NG struggles to consistently estimate correctness for that specific question. Figure 3 provides One-vs-all ROC curves for the three categories in EngSAF: incorrect, partially correct and correct. For the *partially correct* class, we assumed that its scores should lie near the midpoint of the 0–100 scale, and therefore defined the predictor as the absolute distance from 50. This transformation captures how close the model score is to the theoretical midpoint for partial correctness, allowing us to compute a standard ROC curve and derive the AUC for that class. The highest AUC (0.89) is observed when detecting responses labeled as incorrect, suggesting that NodeGrade excels at identifying clearly wrong submissions. Performance remains strong for the fully correct category with an AUC of 0.84, while distinguishing partially correct responses yields a lower but still acceptable AUC of 0.72. This pattern is consistent with the QWK of 0.59 (Table 2), signifying that the model effectively differentiates between clearly correct and clearly incorrect responses, yet finds intermediate correctness levels more challenging to classify.

Figure 4 shows the ROC curve for the SemEval dataset, where the binary target reflects whether a response was considered correct or incorrect. The AUC of 0.81 demonstrates good discriminative power across different decision thresholds, agreeing with the moderate QWK results in Table 2. The ROC curve consistently remains well above the diagonal reference line, underscoring the model's ability to rank correct answers higher than incorrect ones in most cases.

4.2 Comparison of expert and tool

To explore the quality of NG's scoring a comparison between multiple human expert (HE) raters and NG seems useful. Table 4 shows the different ratings that were provided for the student answers

Table 4: Evaluation of student answers by human experts depicting score and classification

	HE1	HE2	HE3	HE4	AV	NG	DS
SA1	10/f	20/f	0/f	30/f	15/f	56.3/f	+41.3
SA2	12/f	30/f	60/p	60/p	40.5/f	70/p	+29.5
SA3	60/p	60/f	100/p	60/p	70/p	58.8/f	-11.2
SA4	75/p	70/p	100/p	100/p	86.3/p	75.4/p	-10.9
SA5	90/p	80/p	80/p	90/p	85/p	72.2/p	-12.8
SA6	91/p	90/p	85/p	100/p	91.5/p	76.4/p	-15.1

SA = Student Answer; HE = Human Expert; AV = Average Human Experts; DS = Difference in Score between AV and NG; p = pass; f = fail

(SA). Each of the three student pairs produced two answers. SA1-4 were produced in response to a task about the strategy pattern, while SA5 and 6 were the results of a task about the decorator pattern. In none of the cases does NG disagree with a clear majority of expert raters considering the classification "pass" or "fail". It is also evident that expert raters in general do not always grade software engineering exercises consistently since for example SA2 only passes for two experts. Looking at SA3 there also seems to be no consensus on which scores represent passing because HE1, HE2 and HE4 give the same score but differ in their classifications. In only one case, SA3, NG gives a failing score while a majority of expert raters, and dependent on this the average, would give the answer a passing score. In reverse only SA2 passes for NG but fails for the majority of experts and on average. There are two cases, SA2 and SA3, NG's classification does not represent the average of human experts. DS varies between +41.3 and -15.3, being the highest for SA1 and SA2, both of which would have probably failed for the majority of human experts, having the lowest average by a considerable margin. SA4 is graded the most similar by experts and NG, with SA3 following close behind. SA6 and SA1 get the highest and lowest score respectively from the HE and NG. Keeping in mind the very small sample size an analysis of outliers or the significance of differences and averages is not possible.

4.3 Qualitative content analysis and observations

Six students worked in pairs on one device with NG, writing answers and getting feedback from it, for about an average of 17 minutes and talked for an average of 5 minutes about NG in their retrospective interview. These interviews represent the data base for the QCA that resulted in four major points.

4.3.1 Usability. NG was described as innovative and praised for its almost human-like interaction. Its simplicity and clarity in operation stood out. Participants noted that the usage was straightforward, with no unnecessary complexity in execution. While some respondents expressed initial reluctance, especially regarding writing tasks, they later acknowledged that engaging with NG encouraged reflective thinking and provided an opportunity to assess the validity of their own reasoning. The user experience appears to shift positively after its first use. On the flipside NG's user-friendliness is hindered by several notable issues identified by the interviewees. First, free-text tasks are perceived as inherently challenging due to the difficulty of identifying precise answers, compounded by technical issues such as spelling errors generated by NG. Second, limited transparency and guidance emerges as a key concern. Participants highlighted the absence of early indications of task constraints, such as character limits, as well as ambiguity regarding whether tasks are intended to mimic practical exam scenarios or serve as open-ended exercises. This lack of clarity makes decisions about the appropriate depth and length of responses more difficult. Finally, problems with task weighting and time management were raised. The absence of clear benchmarks for effort allocation, such as those commonly found in exam settings (e.g., points per minute), creates uncertainty about how much detail is expected in responses.

4.3.2 Feedback mechanisms. The second major point spans the student opinions on three ways of feedback: score, classification and verbal tips. The score system is generally perceived as a valuable and motivating feature by the students. It represents an easy quantitative assessment and enables users to evaluate their current performance clearly, providing a tangible reference point, such as "66 out of 100," to assess their standing, which is highly appreciated. Furthermore the precision of the score, including calculations to one decimal place, is viewed positively as it underlines the extent of necessary improvements, such as the absence of significant aspects or entire sentences. Additionally there is criticism regarding the transparency of the scoring process. Respondents expressed uncertainty about how the score is calculated and what specific actions are required to achieve the maximum score of 100. Despite efforts to improve their responses based on feedback, the path toward achieving full marks often remains unclear. The classification system of "Pass/Fail" is generally perceived as meaningful, particularly in cases where results are close to the threshold for passing. However, in instances where the outcome is clear, such as very low scores, participants regard the classification as less necessary and tend to pay it little attention. A significant issue identified by students is the insufficient clarity regarding the criteria for passing. They expressed a desire for a clear definition of the threshold for "Pass," especially among users with high standards or perfectionistic tendencies. Suggestions include a visual indicator, such as a line in the progress bar, to clearly mark the boundary between passing and failing. Lastly the written feedback provided by NG is largely perceived as helpful and impressive by interviewees. They appreciated the clear and concise suggestions for improvement, which often felt personalized and highlighted aspects relevant to exam preparation. The feedback was seen as a valuable resource for reflection and enhancing the quality of responses. However, inconsistencies were noted. For instance, similar criticisms were repeated for revised answers without clearly indicating what specifically required further improvement. Moreover the perceived iterative structure of the feedback, where new issues are revealed only after each revision, was noticed as frustrating and less efficient. Participants expressed a preference for more comprehensive and conclusive feedback to avoid a step-by-step correction process. Additionally, a lack of alignment between the feedback and the task's level of specificity caused confusion; when general tasks were answered in general terms, feedback unexpectedly requested further specificity. Feedback that heavily emphasized the use of technical terms was

criticized for not adequately accommodating alternative ways to demonstrate understanding. Uncertainty regarding the reliability of NG emerges as a key concern among students. Firstly, it remains unclear to what extent NG accounts for grammatical or spelling errors, leaving its impact on assessments unsure. Secondly, doubts were raised about NG's ability to provide accurate and subject-specific evaluations of user performance. Respondents expressed skepticism about whether the feedback reliably reflects their content knowledge and skills. Lastly, NG's capacity to address significant misunderstandings or errors was questioned. For example, it is not clear whether NG could effectively identify and correct fundamental misinterpretations, as this aspect was not explicitly tested.

4.3.3 Learning process support. The students perceived NG as learning process support, which was the third major point, by combining clear goal-setting with motivational elements. It is portrayed as particularly beneficial for users who are less inclined to write extensively, as it provides guidance on the minimum level of writing work needed to achieve a desired score in exams. Additionally, NG fosters engagement and enthusiasm, with participants highlighting the enjoyment derived from interacting with it as a key motivator in their learning journey. The well-structured nature of the text-based tasks was also praised, with interviewees expressing a willingness to continue using similar tasks in their learning routines.

4.3.4 Future expectations. Regarding the fourth and final major point, students expressed several future expectations for NG, with a strong emphasis on expanding the variety of available tasks. In particular, they desire more free-text and practical tasks that mimic real exam scenarios, which they find invaluable for preparation. A suggestion was made to ensure at least one task per topic to enable more comprehensive practice. Specific ideas include interactive exercises, such as the creation and evaluation of diagrams in software engineering. Furthermore, aligning tasks more closely with exam conditions by providing details on expected answer length or point allocation was seen as a priority. Clearer task formulations were also requested to prevent ambiguity and ensure efficient response construction.

4.3.5 Observations. The major conclusion drawn from the observations of students using NG were the following. Students faced significant challenges in translating their thoughts into coherent written responses, maybe a difficulty exacerbated by working in pairs. While discussions among students often revealed thorough and well-reasoned arguments, these are not consistently reflected in their written submissions. Additionally, students believed that conveying the correct idea should suffice, even without the use of precise technical terminology. They approached the task by meticulously reading and discussing the assignment and feedback, yet often encountered discrepancies between their self-assessment and NG's feedback. They perceived their answers as superior to NG's assessment, which resulted in dismissing NG's suggestions as only partially relevant. Students frequently attempted minimal revisions to test whether these yield higher scores or considered resubmitting the exact response from NG. Students felt that achieving a perfect score would require disproportionate effort. Furthermore, while students attempted one retry, they made no additional revisions. Usability problems were not visible and the students worked seemingly intuitive with NG.

5 Discussion

5.1 Comparison to existing approaches

To address RQ1 and RQ1.1, we compare NG's performance with prior work and public datasets. The authors of [27] report an average Pearson's r of 0.62 on the Mohler dataset, with individual question scores ranging from 0.34 to 0.74. [13] report a Pearson's r of 0.63 on this dataset. The work of [7] presents two approaches: one based on embeddings, which achieves a Pearson's q of 0.557, and a nine-shot approach with context enrichment using a domain-specific corpus, which attains a Pearson's r of 0.525. The works of [17] achieve a Pearson's r of 0.761 on the Mohler dataset using a BERT fine-tuned LSTM model. They also report a Pearson's r of 0.367 for BERT embeddings alone, which employ mean pooling with cosine similarity—a method similar to our approach of using semantic representations from a sentence transformer combined with cosine similarity. Ans2vec, as described in [13], achieves a Pearson's r of 0.63, while our approach attains a Pearson's r of 0.73 using both English and German prompts. This indicates a strong correlation between the model scores and the human ratings, even in comparison to previous studies. Although fine-tuned models appear to yield higher linear correlations, fine-tuning on a single dataset may restrict applicability to broader contexts.

When comparing the work of [27], which explored rubrics for grading responses, our approach obtains a comparable Pearson's r (an average of 0.51 across all questions with the German prompt) on their novel OS dataset relative to their coarse-grained baseline (Pearson's r = 0.415). Their iteratively refined approach achieves a higher correlation (Pearson's r = 0.61), whereas our method is zero-shot in nature. The QWK [6] metric measures the agreement between NG's predictions and human ratings by accounting for both the observed agreement and the agreement expected by chance. QWK values range from −1 (indicating complete inverse agreement) to 1 (indicating perfect agreement), with 0 representing random concordance [6]. In the OS dataset, QWK scores vary by item— for example, Question 3 achieves approximately 0.68 (signifying strong agreement), whereas Question 6 only reaches around 0.20, highlighting challenges with that item. Similarly, a QWK of 0.59 in the EngSAF dataset suggests moderate alignment with expert ratings. Notably, no previous studies have reported QWK benchmarks for these datasets, which underscores the novelty of our evaluation and the need for further research.

On SciEntsBank, [13] report a maximum F1 score of 0.58. According to [18], GPT-4 attains an F1 score of 0.74 on the two-way unseen answers in SciEntsBank and 0.61 on UA two-way in Beetle, yielding a combined F1 score of 0.713. In contrast, by applying threshold optimization aimed at maximizing the F1 score, NG achieves an F1 score of 0.742 on SciEntsBank and 0.724 on Beetle. This result outperforms GPT-4 on the combined SemEval 2013 Task 7 [8] (combined F1 score of 0.738) and demonstrates the enhanced precision of our approach in capturing subtle differences in answer quality. Although selecting the optimal F1 threshold per dataset yields impressive performance, such optimization is impractical for unseen

data and classroom settings. It is therefore recommended that only minimal adjustments be made to the threshold in practical applications, with instructors scaling the score range according to their specific requirements.

5.2 Comparison of expert and tool

While the sample size is very limited and does not allow a statistical analysis there could be a tendency of good or very good answers getting a passing grade but NG being not as gracious in its given point value. On the other hand NG does seem to struggle with grading bad or very bad answers and avoiding low scores. Generally speaking NG seems to grade more around the median and avoid extreme scores. When the results of NG are interpreted sensitively around the median, one can argue that its tendencies align with the human expert ratings addressing RQ2. A bigger pool of student answers and expert ratings is necessary in order to verify these claims for the discipline of software engineering. For teachers there seems to be potential for NG either helping with grading exam answers or regular teacher-student feedback loops. Although since the raters did not know the NG score to avoid a bias, there is no conclusion whether they would be satisfied with NG's work. For students NG could offer relatively realistic practice. It could be argued that NG being not that harsh is beneficial to their motivation since it praises their work but also shows them their mistakes. On the flipside an inflated score from NG could lure students into thinking that they are adequately prepared for the exam, while this is not actually the case. This overconfidence could lead to worse grades. The reluctance from NG to give really high scores for answers the experts deemed to be completely sufficient could lead to two different outcomes. Perfectionist students could lose motivation when trying, and failing, to get a full score but NG's feedback probably does not result in overconfidence. The lower scores could signal a necessity to students to learn more for the following exam, which could show itself in the form of better grades. The results of the QCA and the observations suggest that both scenarios could be realistic. Confidence and self-concept in general are relevant factors for learning and its outcomes [14]. Dedicated further studies into the satisfaction of human raters with NG's performance and the effect of its scores on student learning processes as well as outcomes are needed to prove these different assertions.

5.3 Qualitative Content Analysis

The insights gained based on the QCA of retrospective interviews and the observations, to answer RQ3, underscore NG's potential to combine accessibility with functionality, fostering both ease of use and cognitive engagement. Mentioned issues also collectively highlight significant barriers to effective and confident tool usage, stemming from unclear expectations, insufficient structural support, and technical limitations. In regards to the feedback mechanisms the findings suggest that while the score system is an effective tool for motivation and self-assessment, greater clarity in its calculation and criteria is needed to enhance user understanding and trust. While the written feedback is seen as a strong feature, improvements in consistency, task alignment, and context sensitivity could further enhance its effectiveness. This is in line with other studies arguing a lack of transparency furthers lower trust [26]. The demand for completeness also fits with its high importance for increasing trust [19]. Addressing these concerns through targeted testing and increased transparency could bolster confidence in NG's reliability and enhance its perceived value, with trust levels influencing user acceptance [4]. Overall, NG seems to facilitate a practical and enjoyable approach to exam preparation, making it a valuable resource for learners. By integrating the student suggestions for future development, NG could further enhance its relevance and effectiveness in supporting the learning process through information rich feedback [15]. Including the creation and evaluation of UML diagrams seems to be feasible as well [16]. Qualitative research lacks the ability to prove the degree, intensity and representativeness of these results because of its case based nature. Nevertheless, it gives the opportunity to explore potentially relevant future research questions concerning the wider use of NG.

6 Conclusion

In the context of software engineering education, artificial intelligence and adaptive learning systems, the use of ASAG assists both teachers and students in addressing the challenges associated with the time-intensive yet valuable process of grading free-text and essay-like exercises. The goal of this study was to evaluate NG, an ASAG tool with a focus on performance and usability. To achieve this, we compared its outcomes on public datasets, aligned them with evaluations from human expert raters, and collected feedback from students regarding their experiences with NG. Our results indicate that while NG's performance on public datasets is adequate, there remains room for improvement. For instance, although NG tends to rate student answers on the Mohler and SemEval datasets systematically lower than human raters, its performance on the OS dataset demonstrates excellent operation depending on the question type. In the context of the EngSAF dataset, evidence suggests that NG produced both significantly lower and significantly higher grades compared to human evaluations [7].

Comparison with multiple human expert ratings further suggests that NG struggles with accurately scoring cases at the lower and upper boundary — namely, very good and very poor student answers — although it generally agrees with experts on whether an answer passes or fails in the majority of instances. The comparison also revealed variations in expert grading for software engineering tasks; however, the small sample sizes hinder statistical analysis, and further studies are warranted. The QCA of student interviews and associated observations revealed both positive and negative aspects of NG. For example, students were impressed by the verbal feedback and perceived NG as enhancing motivation and supporting exam preparation. Furthermore, students indicated a desire for greater transparency and improved consistency. These suggestions for future development will be taken into account in the ongoing process of optimizing NG for practical use. A notable limitation of this study is the scarcity of high-quality datasets on which we evaluated our tool, which is compounded by the variability in grading standards or rubrics across sources. Furthermore, the performance of the tool presented in [11] is inherently linked to the quality of

[7] Further data and visualization at: https://github.com/HASKI-RAK/paper/tree/main/NodeGradeEvaluation

the node graph constructed for grading, classification, and feedback generation.

Future research concerned with evaluating NG should address model parameter tuning, explore alternative node configurations, and investigate different weightings between the model output and semantic similarity measures. We did not investigate the concept of Chain-of-Thoughts, as discussed in [21], despite its promising potential demonstrated in recent publications [5]. Similarly, our approach does not incorporate prompt engineering, a limitation also noted in [18]. Furthermore, we did not conduct a cost-benefit analysis or examine the impact of employing various large language models with different hyperparameters.

Issues inherent to manual grading, such as bias [10], continue to affect systems like NG, as the underlying models are typically trained on extensive and diverse textual corpora. Moreover, the limited contextual information available in many public datasets (e.g., SciEntsBank, Beetle, Mohler, EngSAF) poses challenges for approaches that rely on semantic cues to accurately interpret both student responses and reference answers.

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References

[1] Dishank Aggarwal, Pushpak Bhattacharyya, and Bhaskaran Raman. 2024. "I understand why I got this grade": Automatic Short Answer Grading with Feedback. arXiv:2407.12818 [cs.CL] https://arxiv.org/abs/2407.12818

[2] Christine P. Chai. 2023. Comparison of text preprocessing methods. *Natural Language Engineering* 29, 3 (2023), 509–553. doi:10.1017/S1351324922000213

[3] Li-Hsin Chang and Filip Ginter. 2024. Automatic Short Answer Grading for Finnish with ChatGPT. *Proceedings of the AAAI Conference on Artificial Intelligence* 38, 21 (Mar. 2024), 23173–23181. doi:10.1609/aaai.v38i21.30363

[4] Seongyune Choi, Yeonju Jang, and Hyeoncheol Kim. 2022. Influence of Pedagogical Beliefs and Perceived Trust on Teachers' Acceptance of Educational Artificial Intelligence Tools. *International Journal of Human-Computer Interaction* 39 (04 2022), 1–13. doi:10.1080/10447318.2022.2049145

[5] Yucheng Chu, Hang Li, Kaiqi Yang, Harry Shomer, Hui Liu, Yasemin Copur-Gencturk, and Jiliang Tang. 2024. A LLM-Powered Automatic Grading Framework with Human-Level Guidelines Optimization. arXiv:2410.02165 [cs.AI] https://arxiv.org/abs/2410.02165

[6] Jacob Cohen. 1968. Weighted kappa: Nominal scale agreement provision for scaled disagreement or partial credit. *Psychological Bulletin* 70, 4 (1968), 213–220. doi:10.1037/h0026256 Place: US Publisher: American Psychological Association.

[7] Ta Nguyen Binh Duong and Chai Yi Meng. 2024. Automatic Grading of Short Answers Using Large Language Models in Software Engineering Courses. In *2024 IEEE Global Engineering Education Conference (EDUCON)*. IEEE, 3 Park Avenue, New York City, U.S., 1–10. doi:10.1109/EDUCON60312.2024.10578839

[8] Myroslava Dzikovska, Rodney Nielsen, Chris Brew, Claudia Leacock, Danilo Giampiccolo, Luisa Bentivogli, Peter Clark, Ido Dagan, and Hoa Trang Dang. 2013. SemEval-2013 Task 7: The Joint Student Response Analysis and 8th Recognizing Textual Entailment Challenge. In *Second Joint Conference on Lexical and Computational Semantics (*SEM), Volume 2: Proceedings of the Seventh International Workshop on Semantic Evaluation (SemEval 2013)*, Suresh Manandhar and Deniz Yuret (Eds.). Association for Computational Linguistics, Atlanta, Georgia, USA, 263–274. https://aclanthology.org/S13-2045/

[9] Paula Escudeiro, José Bidarra, and Nuno Escudeiro. 2009. Evaluating Educational Software. *Journal of Systemics, Cybernetics and Informatics* 1 (01 2009), 177–182.

[10] Oluwole Fagbohun, Nwaamaka Pearl Iduwe, Mustapha Abdullahi, Adeseye Ifaturoti, and Obinna Nwanna. 2024. Beyond traditional assessment: Exploring the impact of large language models on grading practices. *Journal of Artificial Intelligence, Machine Learning and Data Science* 2, 1 (2024), 1–8. doi:10.51219/jaimld/oluwole-fagbohun/19

[11] David Fischer and Georg Hagel. 2024. Enhancing NLP-Based Educational Assessment: A Node-Based Graph Approach for Analyzing Freeform Student Texts. In *2024 47th MIPRO ICT and Electronics Convention (MIPRO)*. Croatian Society for Information, Communication and Electronic Technology - MIPRO, Jadranski trg 1/II, HR-51000 Rijeka, Croatia, 108–113. doi:10.1109/MIPRO60963.2024.10569607

[12] Rujun Gao, Hillary E Merzdorf, Saira Anwar, M Cynthia Hipwell, and Arun R Srinivasa. 2024. Automatic assessment of text-based responses in post-secondary education: A systematic review. *Computers and Education: Artificial Intelligence* 6 (2024), 100206. doi:10.1016/j.caeai.2024.100206

[13] Wael Hassan Gomaa and Aly Aly Fahmy. 2020. Ans2vec: A Scoring System for Short Answers. In *The International Conference on Advanced Machine Learning Technologies and Applications (AMLTA2019)*, Aboul Ella Hassanien, Ahmad Taher Azar, Tarek Gaber, Roheet Bhatnagar, and Mohamed F. Tolba (Eds.). Springer International Publishing, Cham, 586–595.

[14] John Hattie. 2009. *Visible Learning: A Synthesis of Over 800 Meta-Analyses Relating to Achievement.* Routledge, London. 173–178 pages. doi:10.4324/9780203887332

[15] John Hattie and Helen Timperley. 2007. The Power of Feedback. *Review of Educational Research* 77, 1 (2007), 81–112. doi:10.3102/003465430298487

[16] Florian Huber. 2024. Konzeption und prototypische Entwicklung eines computergestützten Systems zur Lehre von UML-Klassendiagrammen für die Unterstützung der hochschulischen Lehre von Software Engineering. https://epub.uni-regensburg.de/58764/

[17] Soumia Ikiss, Daoudi Najima, Manar Abourezq, and M. Bellafkih. 2024. Improving Automatic Short Answer Scoring Task Through a Hybrid Deep Learning Framework. *International Journal of Advanced Computer Science and Applications* 15 (08 2024). doi:10.14569/IJACSA.2024.01508105

[18] Gerd Kortemeyer. 2024. Performance of the pre-trained large language model GPT-4 on automated short answer grading. *Discover Artificial Intelligence* 4, 1 (2024), 47. doi:10.1007/s44163-024-00147-y

[19] Todd Kulesza, Simone Stumpf, Margaret Burnett, Sherry Yang, Irwin Kwan, and Weng-Keen Wong. 2013. Too much, too little, or just right? Ways explanations impact end users' mental models. In *2013 IEEE Symposium on Visual Languages and Human Centric Computing*. IEEE, 3 Park Avenue, New York City, U.S., 3–10. doi:10.1109/VLHCC.2013.6645235

[20] Viet Dac Lai, Nghia Trung Ngo, Amir Pouran Ben Veyseh, Hieu Man, Franck Dernoncourt, Trung Bui, and Thien Huu Nguyen. 2023. ChatGPT Beyond English: Towards a Comprehensive Evaluation of Large Language Models in Multilingual Learning. arXiv:2304.05613 [cs.CL] https://arxiv.org/abs/2304.05613

[21] Gyeong-Geon Lee, Ehsan Latif, Xuansheng Wu, Ninghao Liu, and Xiaoming Zhai. 2024. Applying large language models and chain-of-thought for automatic scoring. *Computers and Education: Artificial Intelligence* 6 (2024), 100213. doi:10.1016/j.caeai.2024.100213

[22] Florence Martin, Yan Chen, Robert Moore, and Carl Westine. 2020. Systematic review of adaptive learning research designs, context, strategies, and technologies from 2009 to 2018. *Educational Technology Research and Development* 68 (06 2020), 1903–1929. doi:10.1007/s11423-020-09793-2

[23] Philipp Mayring. 2014. *Qualitative content analysis - theoretical foundation, basic procedures and software solution.* Mayring, Philipp, Klagenfurt, Austria.

[24] Michael Mohler, Razvan Bunescu, and Rada Mihalcea. 2011. Learning to Grade Short Answer Questions using Semantic Similarity Measures and Dependency Graph Alignments. In *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies*, Dekang Lin, Yuji Matsumoto, and Rada Mihalcea (Eds.). Association for Computational Linguistics, Portland, Oregon, USA, 752–762. https://aclanthology.org/P11-1076/

[25] Michael Patton. 2002. *Qualitative Research And Evaluation Methods* (3 ed.). Sage Publications, Thousand Oaks, London.

[26] Dimitrios Tzimas and Stavros N. Demetriadis. 2021. Ethical issues in learning analytics: a review of the field. *Educational Technology Research and Development* 69 (2021), 1101 – 1133.

[27] Wenjing Xie, Juxin Niu, Chun Jason Xue, and Nan Guan. 2024. Grade Like a Human: Rethinking Automated Assessment with Large Language Models. doi:10.48550/arxiv.2405.19694 arXiv:2405.19694 [cs.AI]

[28] Yuyun, Andi Djalal Latief, Tri Sampurno, Hazriani, Andriansyah Oktafiandi Arisha, and Mushaf. 2023. Next Sentence Prediction: The Impact of Preprocessing Techniques in Deep Learning. In *2023 International Conference on Computer, Control, Informatics and its Applications (IC3INA)*. IEEE, 3 Park Avenue, New York City, U.S., 274–278. doi:10.1109/IC3INA60834.2023.10285805

[29] Olaf Zawacki-Richter, Victoria Marín, Melissa Bond, and Franziska Gouverneur. 2019. Systematic review of research on artificial intelligence applications in higher education -where are the educators? *International Journal of Educational Technology in Higher Education* 16 (10 2019), 1–27. doi:10.1186/s41239-019-0171-0

[30] Štefan Karolčík, Elena Čipková, Milan Veselský, and Helena Hrubišková. 2015. The Comprehensive Evaluation of Electronic Learning Tools and Educational Software (CEELTES). *Informatics in Education* 14, 2 (2015), 243–264. doi:10.15388/infedu.2015.14